

Day 7 1/12/2025

→ while loops

It is used when you do not know in advance how many times the loop needs to run.

* It continues executing until the condition become false

Ex:- $i = 1$

while $i \leq 5$:

 print(i)

$i += 1$

1) what are the loop statements?

⇒ Loop statements are "control structure" that allow a set of instruction to be executed repeatedly until a certain condition is met

Types of loops statement

1. Break

2. continue

3. pass

1. Break:- used to exit / terminate the loop immediately

Ex:- for i in range (1, 11, 1):

 if ($i == 5$):

 break

 else:

 print(i)

output :- 1

2

3

4

2. continue statement:-

It is used to skip the current iteration of the loop

```
Ex:- for i in range (1, 11, 1):  
    if (i == 5):  
        continue  
    else:  
        print(i)
```

3. pass statement:-

It purely does nothing

```
Ex:- for i in range (1, 11, 1):  
    if (i == 5):  
        continue pass  
    else:  
        print(i)
```

Example problem

1) write a program to check the given number is prime number or not.

=> num = int(input("enter a number"))

```
for i in range (2, num, 1):  
    if num % i == 0:  
        print(num, "its not a prime no")  
    else:  
        print(num, "its a prime no")
```


Q2 write a program to count no of natural no, even no, odd no from the given range

```
=> num = int(input("enter a number"))  
num2 = int(input("enter a number"))
```

```
natural = 0
```

```
even = 0
```

```
odd = 0
```

```
for i in range(num, num2+1, 1):  
    natural = natural + 1
```

```
    if i % 2 == 0:  
        even = even + 1
```

```
    else:  
        odd = odd + 1
```

```
print("natural number=", natural)
```

```
print("even number=", even)
```

```
print("odd number=", odd)
```